

The round-three combination

Draft, October 8, 2026

This note assembles the conditional witness

$$T(n) = O(n(\log n)^{1-\kappa}), \quad \kappa = \frac{13086957581}{3125 \cdot 10^{12}} \approx 4.1878 \cdot 10^{-6} > 2^{-18},$$

from four components. It points to where each one is proved or checked.

1 Components

The stack. Ideas A–E (`stack-notes.tex`, `segmented-inverse.tex`, `fine-field-lemma.tex`) drive ϵ toward 1. The binding row is then the simultaneous butterflies, $\epsilon(1 - \lambda')$, with $\lambda' > \max(\tau, \sigma, \chi, \text{leaf})$. The reservations force $\epsilon < 1/(2 - \lambda')$, so the supremum of κ is $\min(a_b, (1 - \beta)a_c)/(1 + \cdot)$.

Bit side: the batched two-stage interchange. This is the two-stage bit motif with partial-swap frames, with the batched recursion extended to partial-swap frames (`partial-swap-batching.tex`, its Lemma 1). At $h = 32$, $m = 1024$, it certifies $a_b = 22157/(5 \cdot 10^9) \approx 4.43 \cdot 10^{-6}$. This side does not bind.

Complex side: PR #7's network, fully batched. PR #7's complex network at $h = 28$ ($m = 21952$), evaluated under PR #15's full batching: every nonzero residual edge, of every rank, is one whole-residual child call. We rebuilt the residual histogram from our own producer, labels and role compilation (`independent/complex-network/fullbatch_hist.py`) and certified the saving by exact bisection on the moment equation with certified upper bounds on $\ln$ (`fullbatch_cert.py`). This gives $a_c \geq 1048009/(2.5 \cdot 10^{11}) \approx 4.192 \cdot 10^{-6}$ (PR #15 reports $4191487/10^{12}$ from its own labels), and this side binds.

Guard. The coefficient-depth guard needs no path-topology assumption. Every child call of the batched recursion acts on at most $(m - 1)/m$ of its parent’s axes, and a node has at most s_c children in sequence. So the dependency depth is at most d^{C_1} with $C_1 = 1 + m_c \ln s_c$, and Idea D takes x with $\epsilon C_1 < 1 + x$ (`scripts/certificate_round3.py`, guard mode `crude`). Any finite C_1 gives the same κ , since x does not enter it; with $x \approx 8.4 \cdot 10^5$ the packed-product and record-regime rows hold for sufficiently large L . The witness is stated with this guard only.

2 Certificate

`scripts/certificate_round3.py` evaluates the rows and constraints exactly for any pair (a_b, a_c) . With $a_b = 22157/(5 \cdot 10^9)$, $a_c = 1048009/(2.5 \cdot 10^{11})$, $\beta = 1/1000$ and the crude guard:

- every constraint is strict;
- the binding row is the simultaneous butterflies;
- $\kappa = 13086957581/(3.125 \cdot 10^{15})$.

`independent/complex-network/fullbatch_hist.py` 28 followed by `fullbatch_cert.py` recomputes a_c and κ . Our histogram differs from PR #15’s in the smallest ranks, with the same total s_c , and certifies at least its value.

3 What this rests on

- The pinned upstream manuscript and its interfaces.
- Colkitt’s compact controls.
- PR #5’s chirped correlation, with PR #10’s correction of its Gaussian input error.
- PR #7’s ternary producer and complex network.
- PR #10’s batched recursion and controlled projector basis; we checked its edge forms and found basis witnesses at $h = 8$ to 14.
- PR #13’s third whole-residual class and PR #15’s full batching of every residual rank, with an orthonormal residual basis on every edge.
- Paureel’s two-stage motif.

Our own components:

- Ideas A–E;
- the partial-swap batching lemma;
- an independent rebuild and certification of the fully batched complex saving;
- the topology-free guard.